



CS 431/631 Introduction to Big Data

Big Data Overview

Some slides are borrowed from John Canny, Mohamed Eltabakh, et. al.

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What is Big Data?

BIG Data Definition



Volume of data created, captured, copied, and consumed worldwide



The volume of data generated, consumed, copied, and stored is projected to exceed 180 zettabytes by 2025

Data VS Computing Power --> Big Data

(Graph 1)

WHAT'S A ZETTABYTE?

1 kilobyte	1,000,000,000,000,000,000
1 megabyte	1,000,000,000,000,000,000,000
1 gigabyte	1,000,000,000,000,000,000,000,000
1 terabyte	1,000,000,000,000,000,000,000,000,000
1 petabyte	1,000,000,000,000,000,000,000,000,000,000
1 exabyte	1,000,000,000,000,000,000,000,000,000,000,000
1 zettabyte	1,000,000,000,000,000,000,000,000,000,000,000,000



Source: statista.com

firstsiteguide.com

Big data happens when the data you have to process is bigger than what you can process currently

*Data + Computing Power + Purpose
--> Big Data*



Big data is when we can handle data fast enough to make a difference

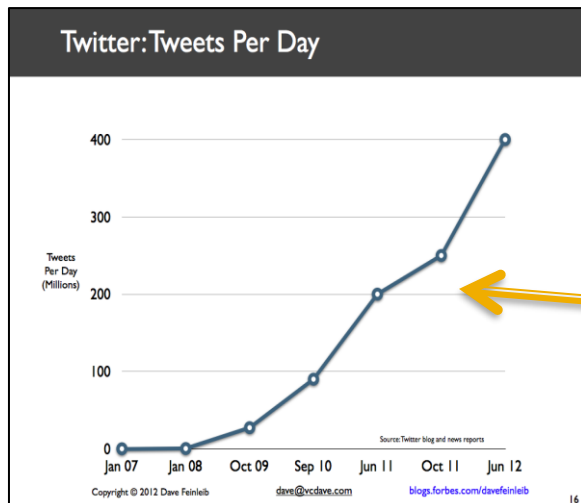
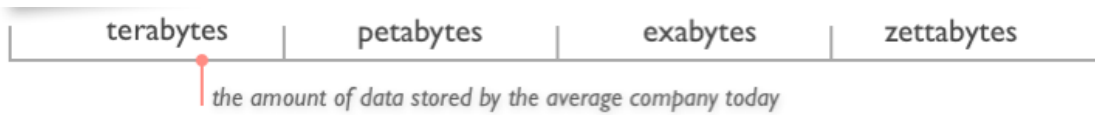
Source: Arnon Rotem-Gal-Oz's big data talk.

Picture: Stephen H. Kaisler, Frank Armour, Alberto Espinosa and William H. Money

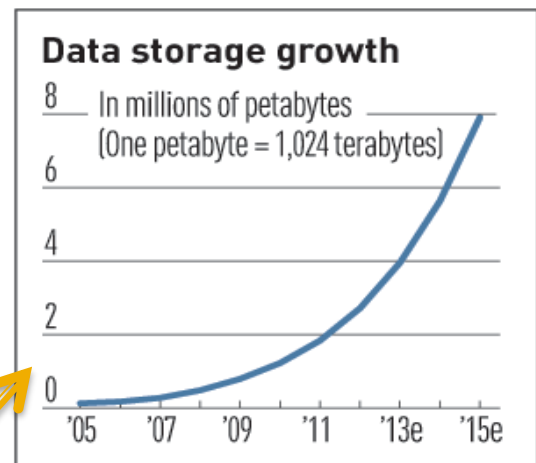
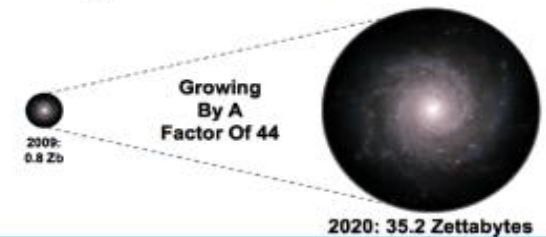
Characteristics of Big Data:

1-Scale (Volume)

- **Data Volume**
 - 44x increase from 2009 to 2020
 - From 0.8 zettabytes to 35zb
- Data volume is increasing exponentially



The Digital Universe 2009-2020



Exponential increase in collected/generated data

Characteristics of Big Data:

2-Speed (Velocity)

- Data is generated fast and needs to be processed fast
- Online data analytics
- Late decisions → missing opportunities



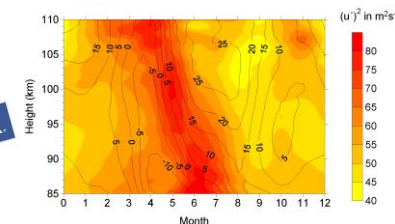
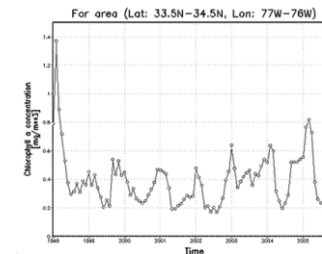
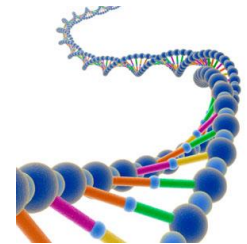
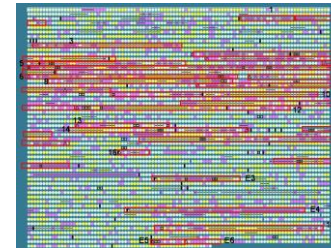
■ Examples

- **E-Promotions:** Based on your current location, your purchase history, what you like → send promotions right now for store next to you
- **Healthcare monitoring:** sensors monitoring your activities and body → any abnormal measurements require immediate reaction

Characteristics of Big Data:

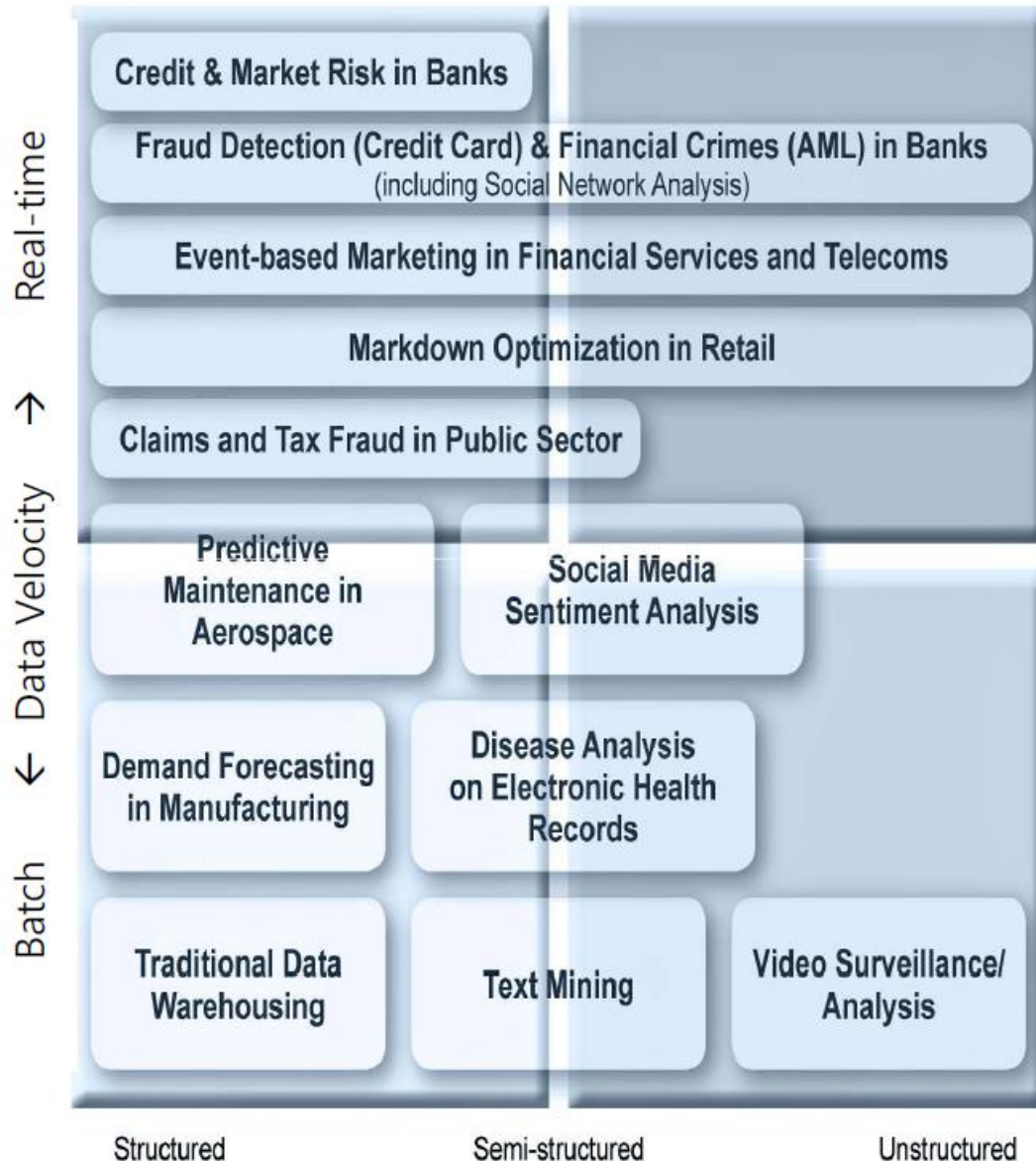
3-Complexity (Variety)

- Various formats, types, and structures
- Text, numerical, images, audio, video, sequences, time series, social media data, multi-dim arrays, etc.
- Static data vs. streaming data
- A single application can be generating/collecting many types of data



To extract knowledge → all these types of data need to be linked together (data fusion)

The Structure of Big Data



- **Structured**
 - Most traditional data sources
- **Semi-structured**
 - Many sources of big data
- **Unstructured**
 - Video data, audio data

Another Way to Look at Data

Quantitative Data

- Measurable
- Collected through measuring things that have a fixed reality
- Close ended

Qualitative Data

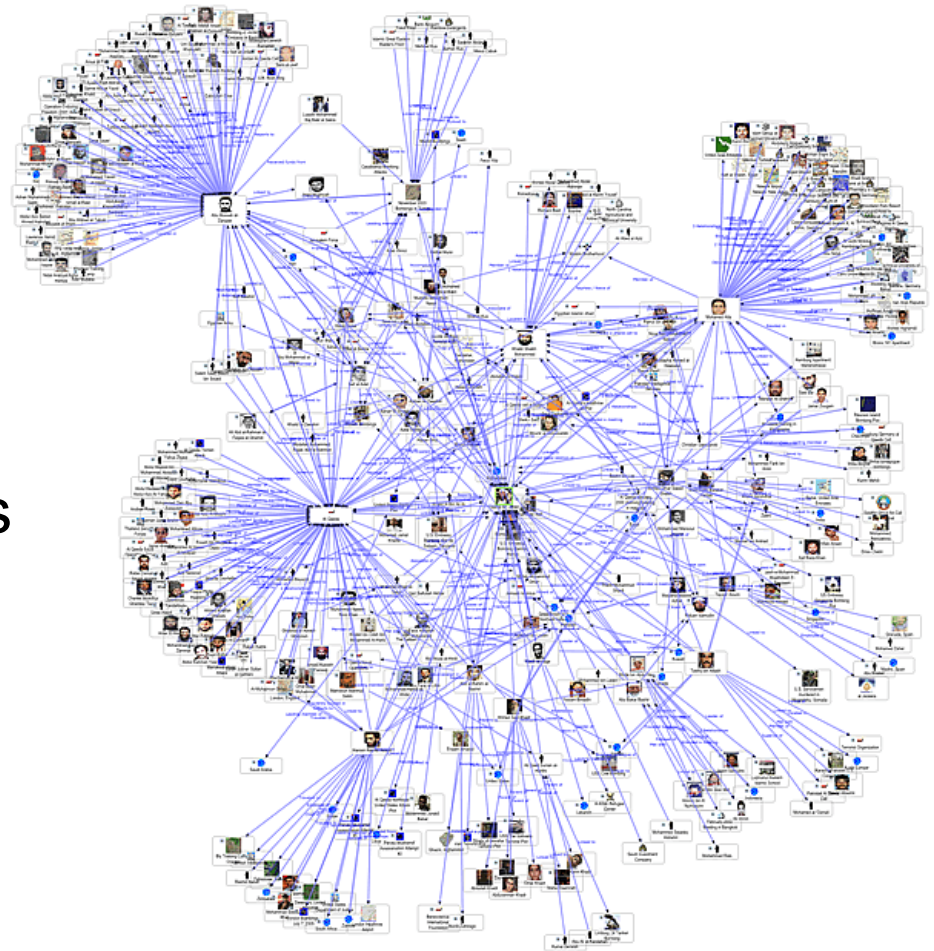
- Descriptive
- Collected through observation, field work, focus groups, interviews, recording or filming conversations
- Open ended

Graph Data

Lots of interesting data has a graph structure:

- Social networks
- Communication networks
- Computer Networks
- Road networks
- Citations
- Collaborations/Relationships
- ...

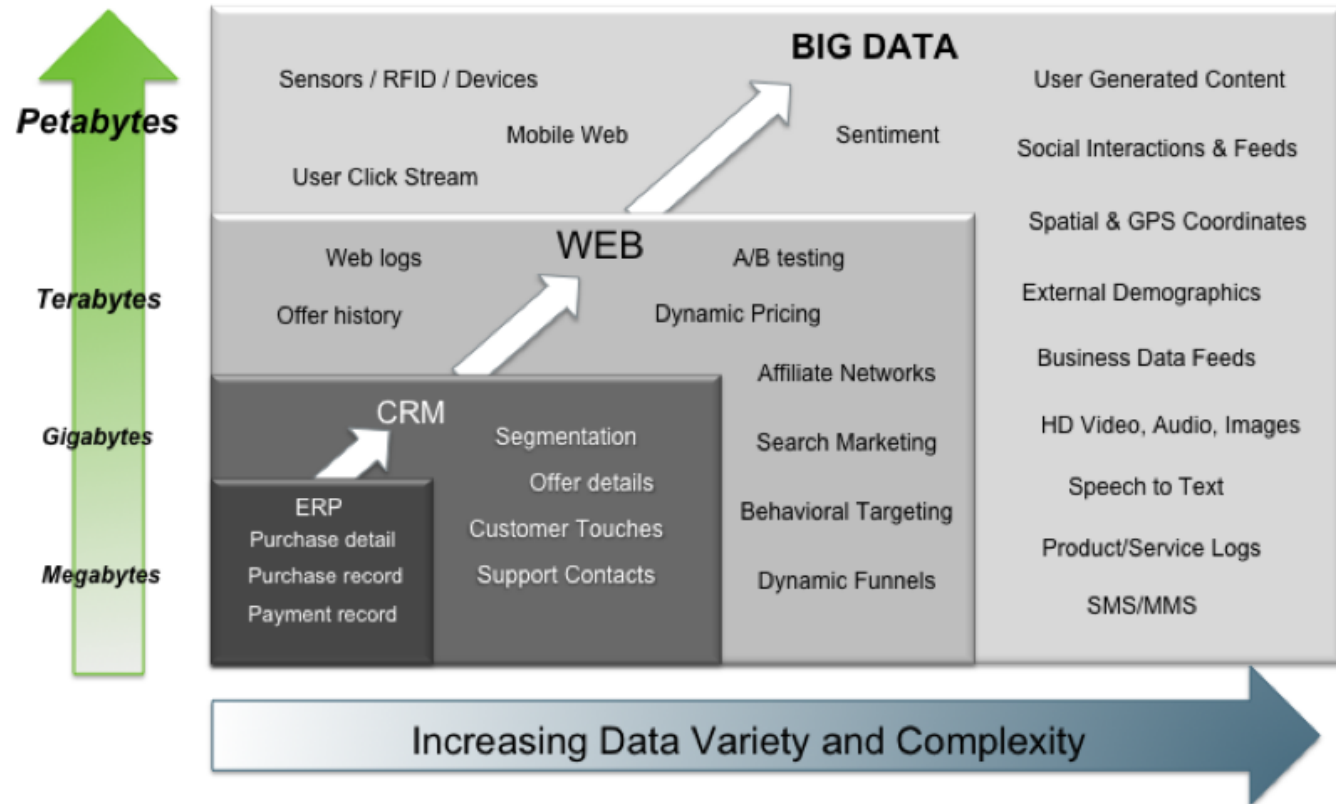
Some of these graphs can get quite large (e.g., Facebook* user graph)



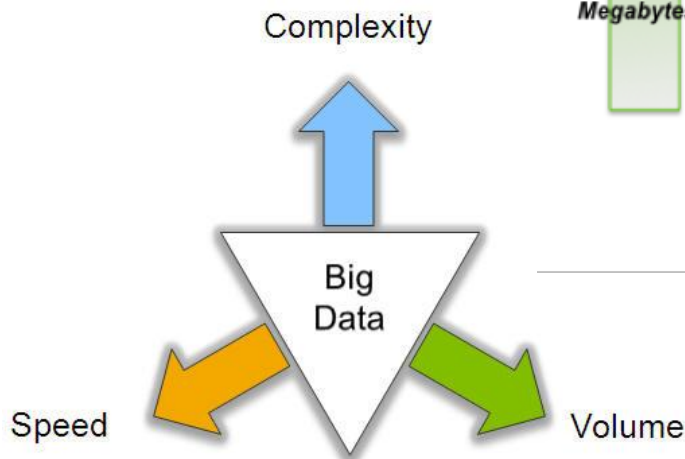
Big Data: 3V's



Big Data = Transactions + Interactions + Observations

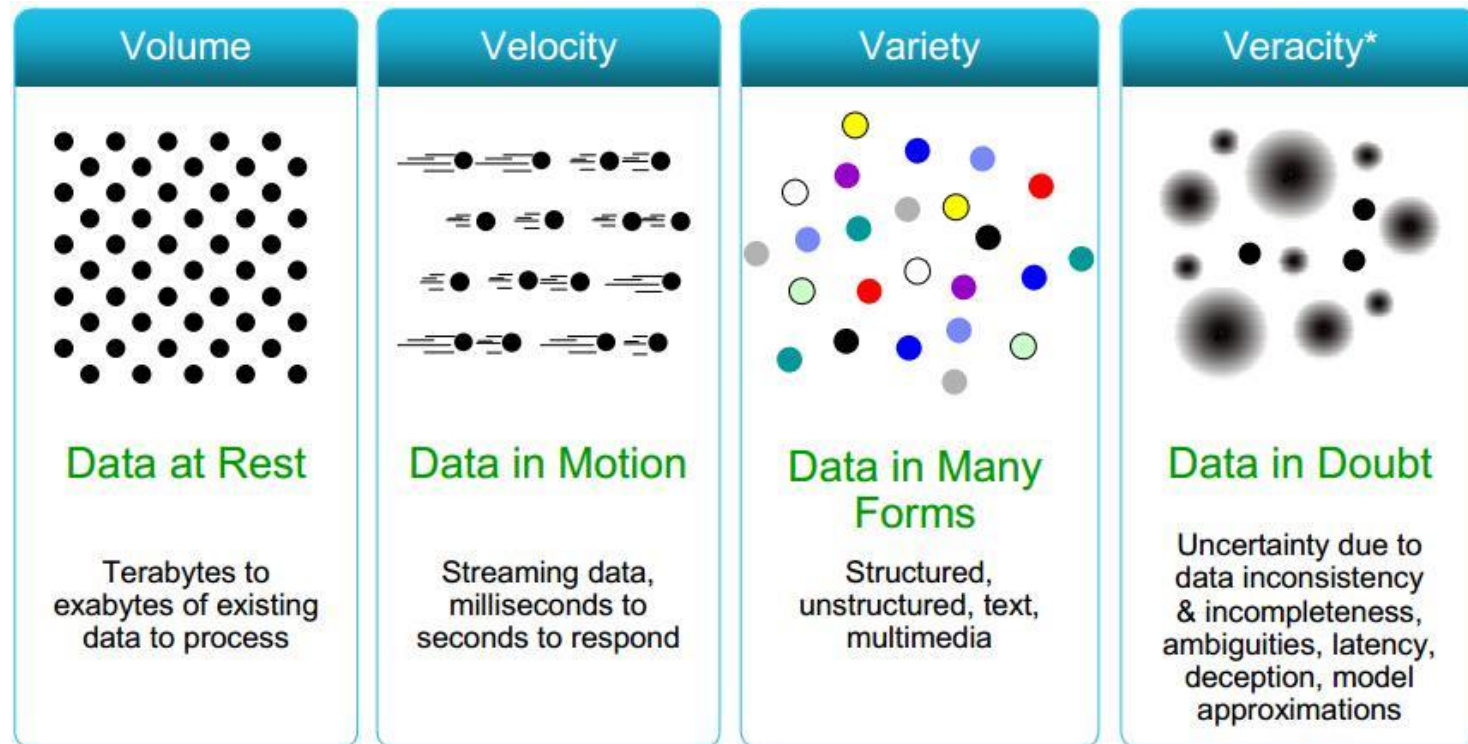


Source: Contents of above graphic created in partnership with Teradata, Inc.

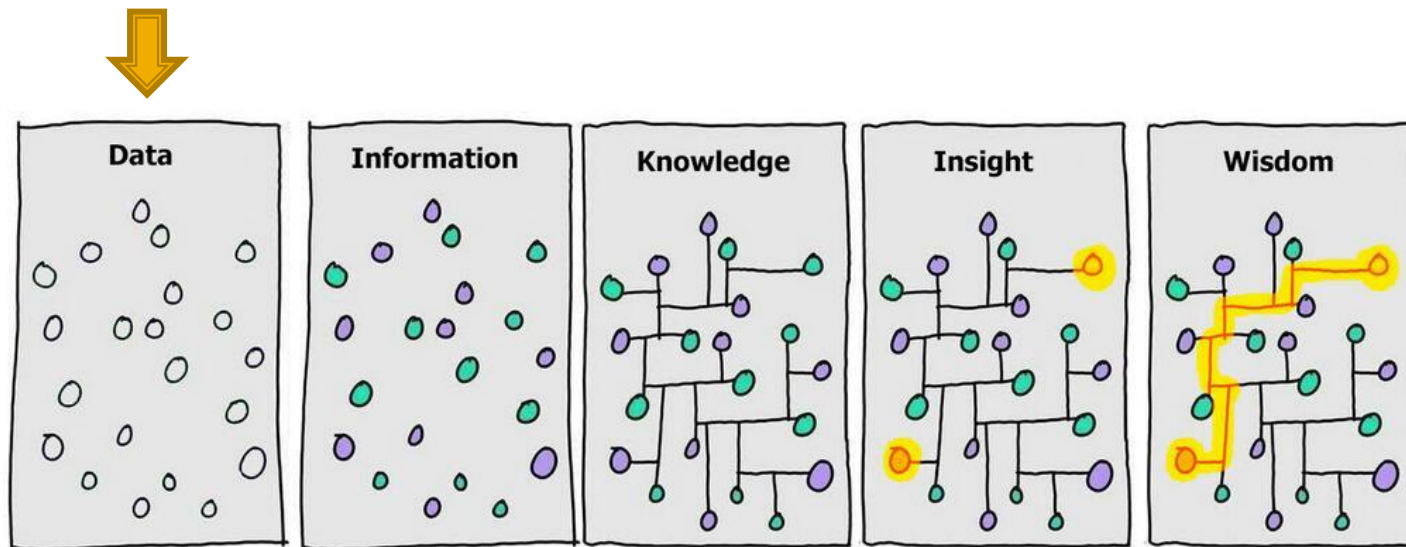


ERP: Enterprise resource planning
CRM: Customer relationship management

Some Make it 4V's

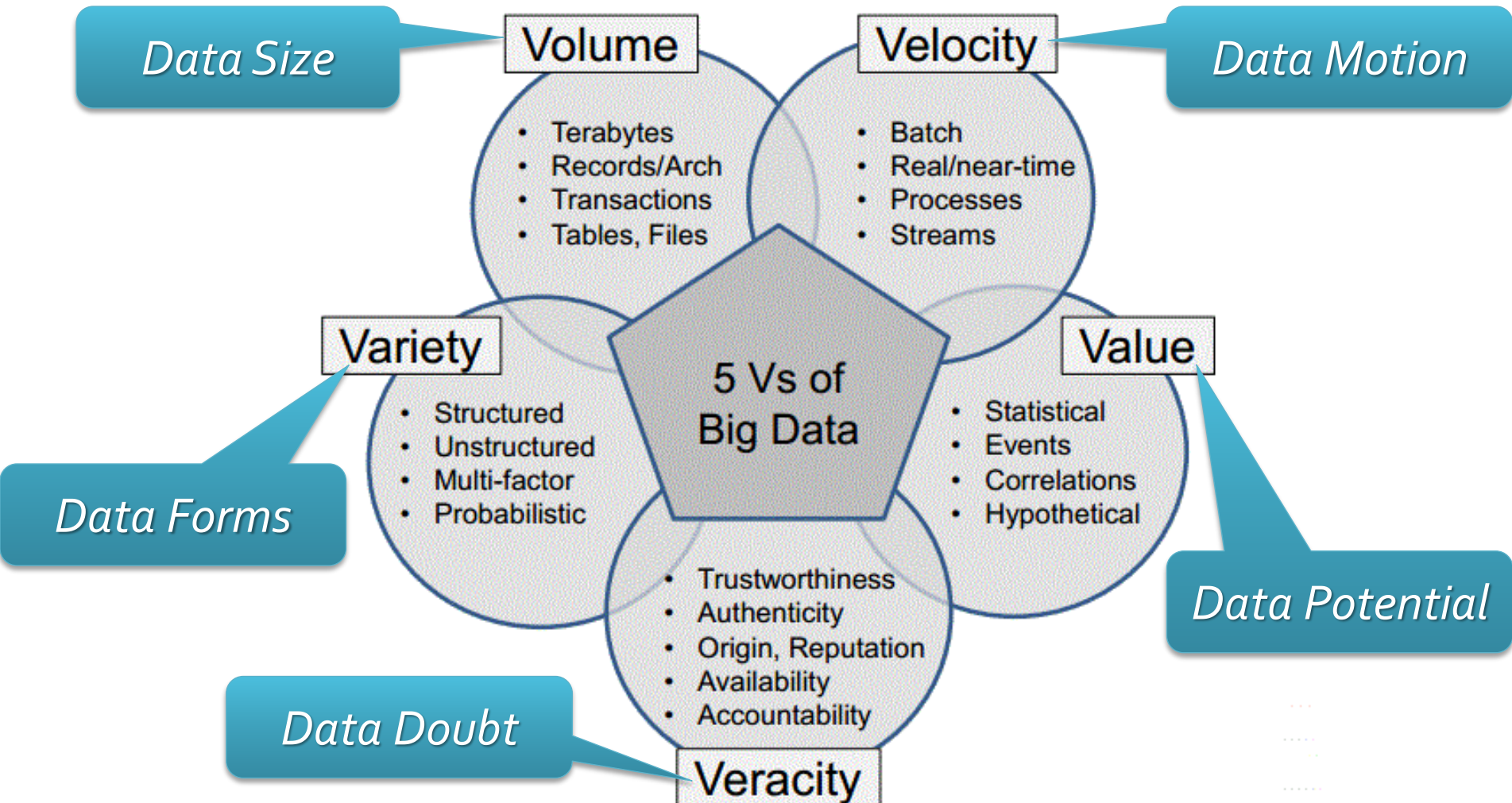


The Information Continuum



Cartoon by [David Somerville](#), based on a two pane version by [Hugh McLeod](#)

Others Make it 5V's



Who's Generating Big Data



Who's Generating Big Data

User Generated (Web & Mobile)



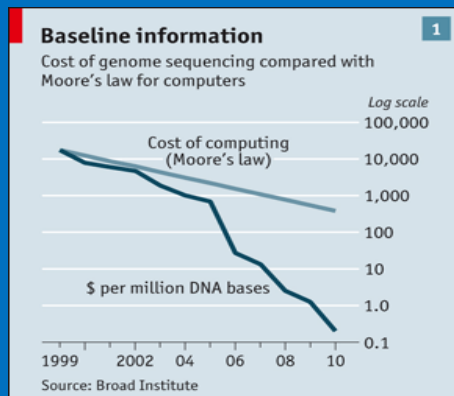
- **Prevalence** – the proportion of the population with a disease at a particular point in time
- **Incidence** – the number of new cases of a disease in a particular population over a defined period of time
- **Prevalence** = **Incidence** x **Duration**

10

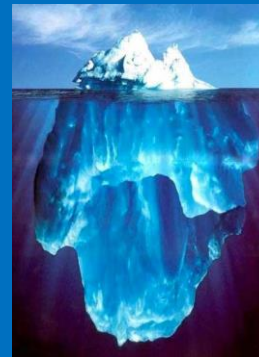
Internet of Things / M2M



Health/Scientific Computing

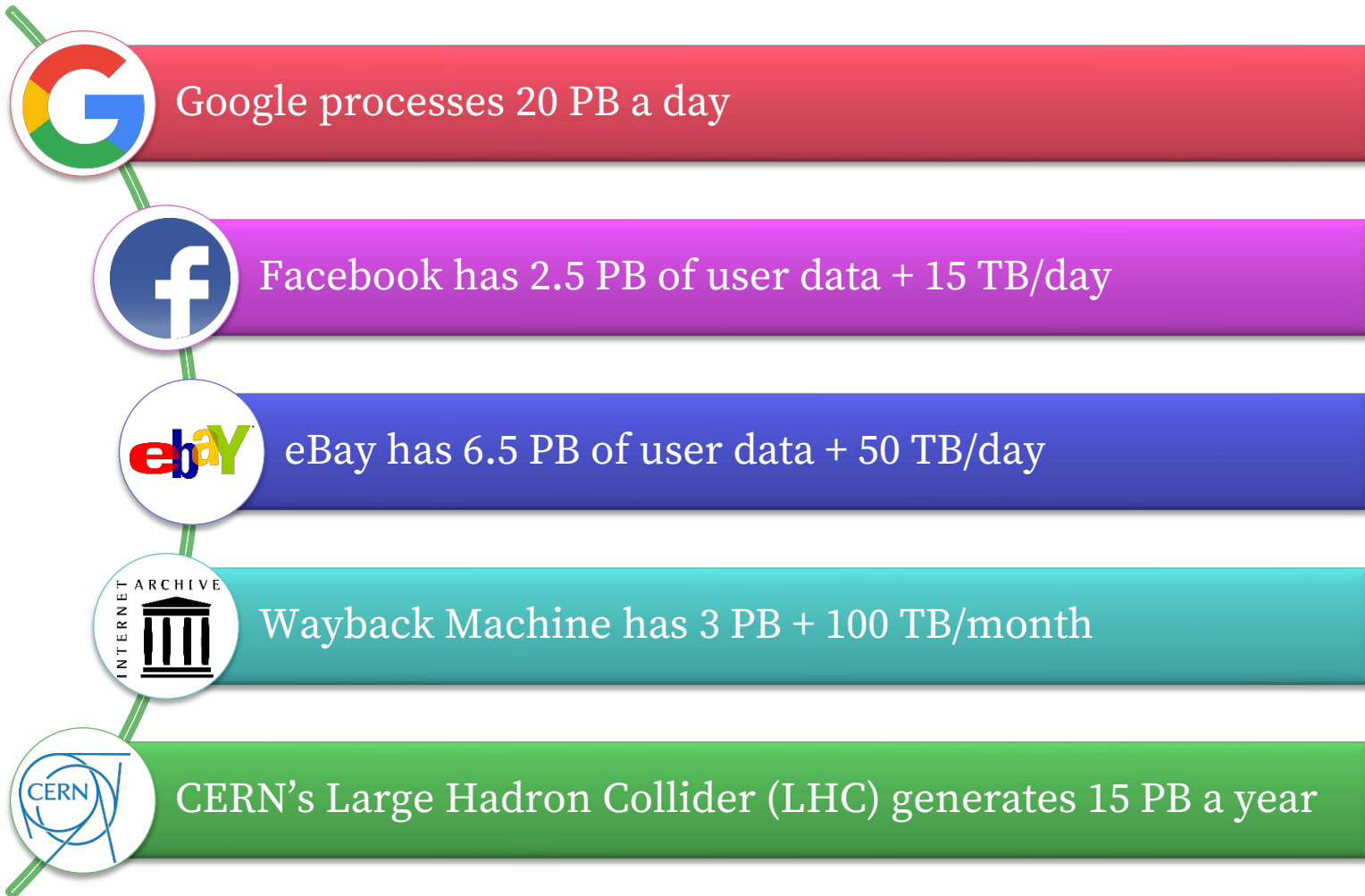


It's All Happening On-line



- Every:
 - Click
 - Ad impression
 - Billing event
 - Fast Forward, pause, ...
 - Server request
 - Transaction
 - Network message
 - Fault ...

Who's Generating Big Data



Who's Generating Big Data



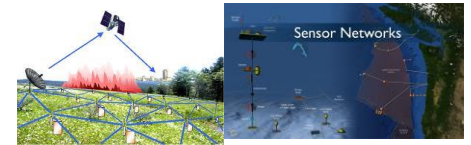
Social media and networks
(all of us are generating data)



Scientific instruments
(collecting all sorts of data)



Mobile devices
(tracking all objects all the time)



Sensor technology and networks
(measuring all kinds of data)

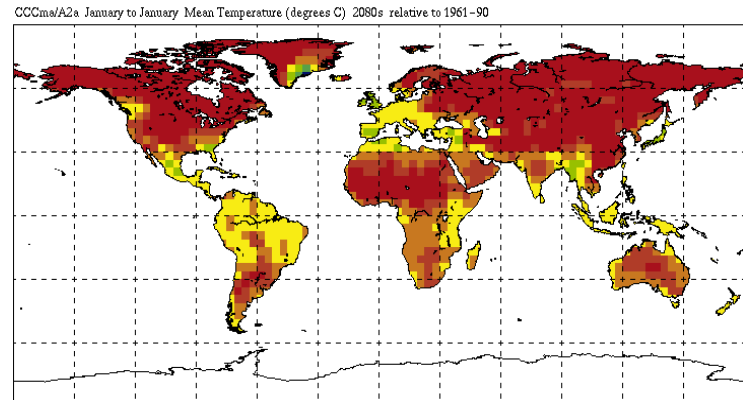
- The progress and innovation is no longer hindered by the ability to collect raw data
- But, by the ability to manage, analyze, summarize, visualize, and discover knowledge from the collected data in a timely manner and in a scalable fashion

What can you do with the data?

Great Opportunities to Solve Society's Major Problems



Improving health care and reducing costs



Predicting the impact of climate change

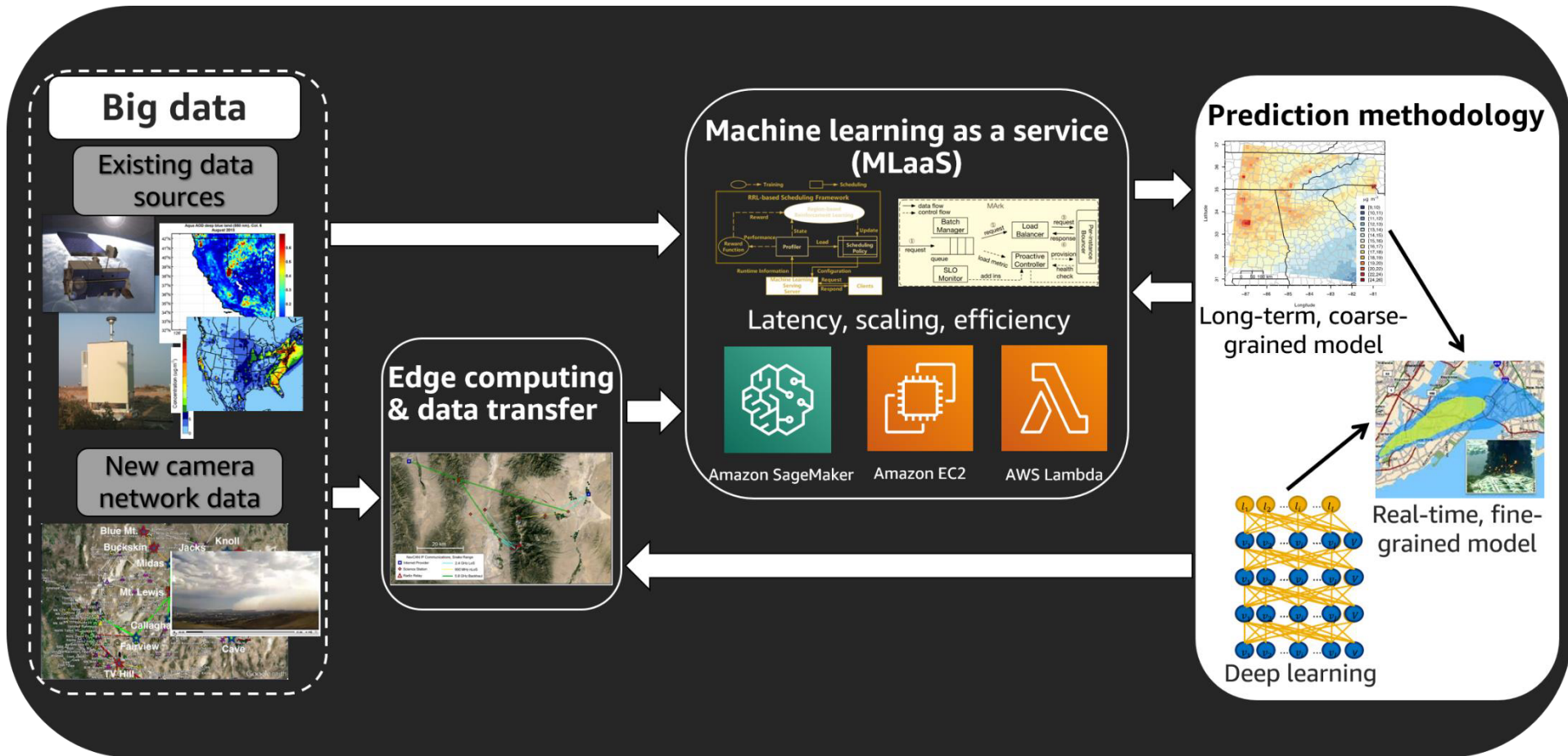


Finding alternative/ green energy sources



Reducing hunger and poverty by increasing agriculture production

What can you do with the data?

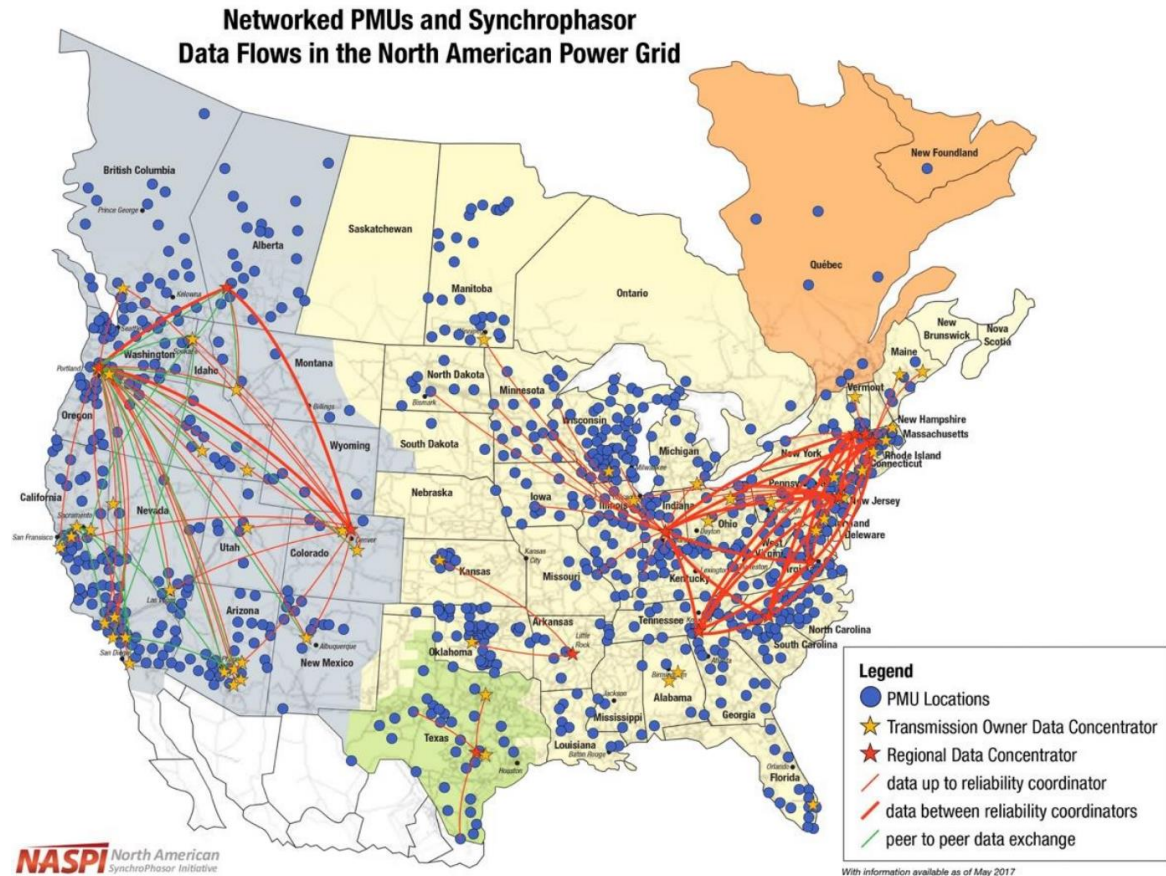


NSF BIGDATA: IA: Collaborative Research: Protecting Yourself from Wildfire Smoke: Big Data Driven Adaptive Air Quality Prediction Methodologies, Lei Yang (Lead-PI), 2018-2023, \$1,401,274 (\$1,042,676 from NSF, \$358,571.34 from AWS)

What can you do with the data?



- Over 2500 networked PMUs
- High level of situational awareness E.g., real-time event detection, classification



DOE: A Robust Event Diagnostics Platform: Integrating Tensor Analytics and Machine Learning Into Real-time Grid Monitoring, Lei Yang (Lead-PI), 2019-2022, \$746,202

Big Data Applications



Big Data Applications

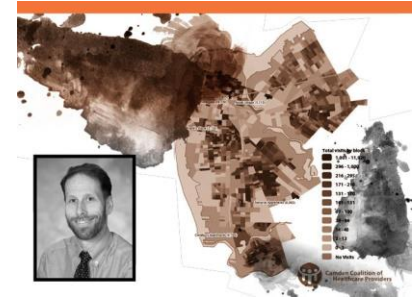
Big Data Analytics: Industry Examples

- 1 Health Care
 - Reducing Cost of Care
- 2 Public Services
 - Preventing Pandemics
- 3 Life Sciences
 - Genomic Mapping
- 4 IT Infrastructure
 - Unstructured Data Analysis
- 5 Online Services
 - Social Media for Professionals



Big Data Applications: Healthcare

1 Big Data Analytics: *Healthcare*



Situation

- Poor police response and problems with medical care, triggered by shooting of a Rutgers student
- The event drove local doctor to map crime data and examine local health care

Use of Big Data

- Dr. Jeffrey Brenner generated his own crime maps from medical billing records of 3 hospitals

Key Outcomes

- City hospitals & ER's provided expensive care, low quality care
- Reduced hospital costs by 56% by realizing that 80% of city's medical costs came from 13% of its residents, mainly low-income or elderly
- Now offers preventative care over the phone or through home visits

Big Data Applications: Public Services

2 Big Data Analytics: *Public Services*



Situation

- Threat of global pandemics has increased exponentially
- Pandemics spreads at faster rates, more resistant to antibiotics

Use of Big Data

- Created a network of viral listening posts
- Combines data from viral discovery in the field, research in disease hotspots, and social media trends
- Using Big Data to make accurate predications on spread of new pandemics

Key Outcomes

- Identified a fifth form of human malaria, including its origin
- Identified why efforts failed to control swine flu
- Proposing more proactive approaches to preventing outbreaks

Big Data Applications: Public Services

COVID-19

Global: <https://gisanddata.maps.arcgis.com/apps/opsdashboard/index.html#/bda7594740fd40299423467b48e9ecf6>

Reproduction Number (R):

https://en.wikipedia.org/wiki/Basic_reproduction_number

If $R_t > 1$, the virus will spread out and the disease will become an epidemic;

if $R_t = 1$, the virus will spread locally and the disease is endemic;

if $R_t < 1$, the virus will stop spreading and the disease will disappear eventually.

Washoe County: <https://gis.washoecounty.us/ago/Host?id=COVID19Dashboard>

Vaccines: <https://www.bloomberg.com/features/2020-coronavirus-drug-vaccine-status/>

(e.g., Moderna's mRNA-based vaccine development is powered by AWS)

Big Data Applications: Life Sciences

3 Big Data Analytics: *Life Sciences*



Situation

- Broad Institute (MIT & Harvard) mapping the Human Genome

Use of Big Data

- In 13 yrs, mapped 3 billion genetic base pairs; 8 petabytes
- Developed 30+ software packages, now shared publicly, along with the genomic data

Key Outcomes

- Using genetic mappings to identify cellular mutations causing cancer and other serious diseases
- Innovating how genomic research informs new pharmaceutical drugs

Big Data Applications: IT Infrastructure

4 Big Data Analytics: *IT Infrastructure*



Situation

- Explosion of unstructured data required new technology to analyze quickly, and efficiently

Use of Big Data

- Doug Cutting created Hadoop to divide large processing tasks into smaller tasks across many computers
- Analyzes social media data generated by hundreds of thousands of users

Key Outcomes

- New York Times used Hadoop to transform its entire public archive, from 1851 to 1922, into 11 million PDF files in 24 hrs
- Applications range from social media, sentiment analysis, wartime chatter, natural language processing

Big Data Applications: Online Services

5 Big Data Analytics: *Online Services*



Situation

- Opportunity to create social media space for professionals

Use of Big Data

- Collects and analyzes data from over 100 million users
- Adding 1 million new users per week

Key Outcomes

- LinkedIn Skills, InMaps, Job Recommendations, Recruiting
- Established a diverse data scientist group, as founder believes this is the start of Big Data revolution

Big Data Applications

elections2012

Live results President Senate House Governor Choose your

Numbers nerd Nate Silver's forecasts prove all right on election night

FiveThirtyEight blogger predicted the outcome in all 50 states, assuming Barack Obama's Florida victory is confirmed

Luke Harding

guardian.co.uk, Wednesday 7 November 2012 10.45 EST



the signal and the
and the noise and
the noise and the
noise and the no
why most noise a
predictions fail
but some don't n
and the noise and
the noise and the
nate silver noise
noise and the no

ELECTION 2020

MEDIA MONITOR

Powered by RavenPack



Big Data Applications

How Formula 1 Teams Use Big Data to Win

by Stacey Higginbotham @gigastacey NOVEMBER 12, 2015, 7:15 AM EST

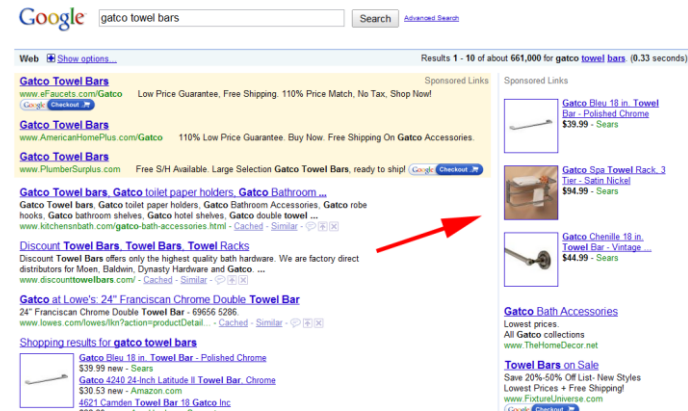


The popular auto racing series also serves as a proving ground for sensors, software, and data-driven decisions.

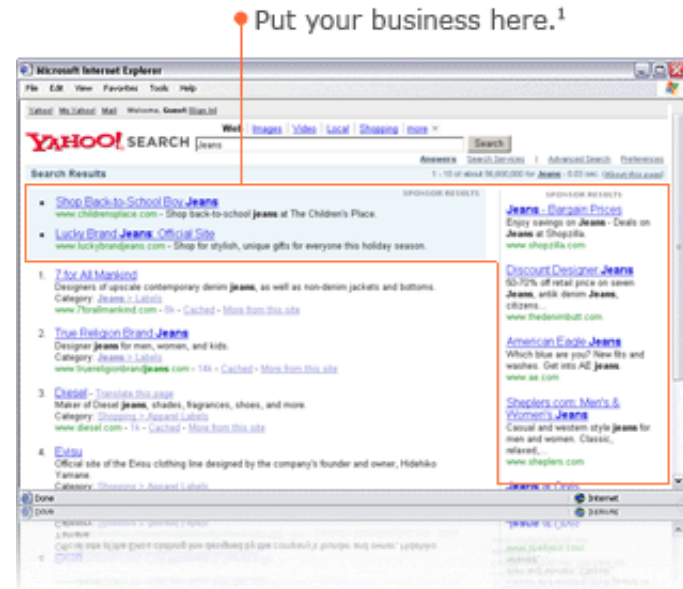
Daniel Ricciardo's car at Infiniti-Redbull Racing's garage at the Circuit of the Americas track in Austin, TX

Article from
FORTUNE

1998 – sponsored search



Overture

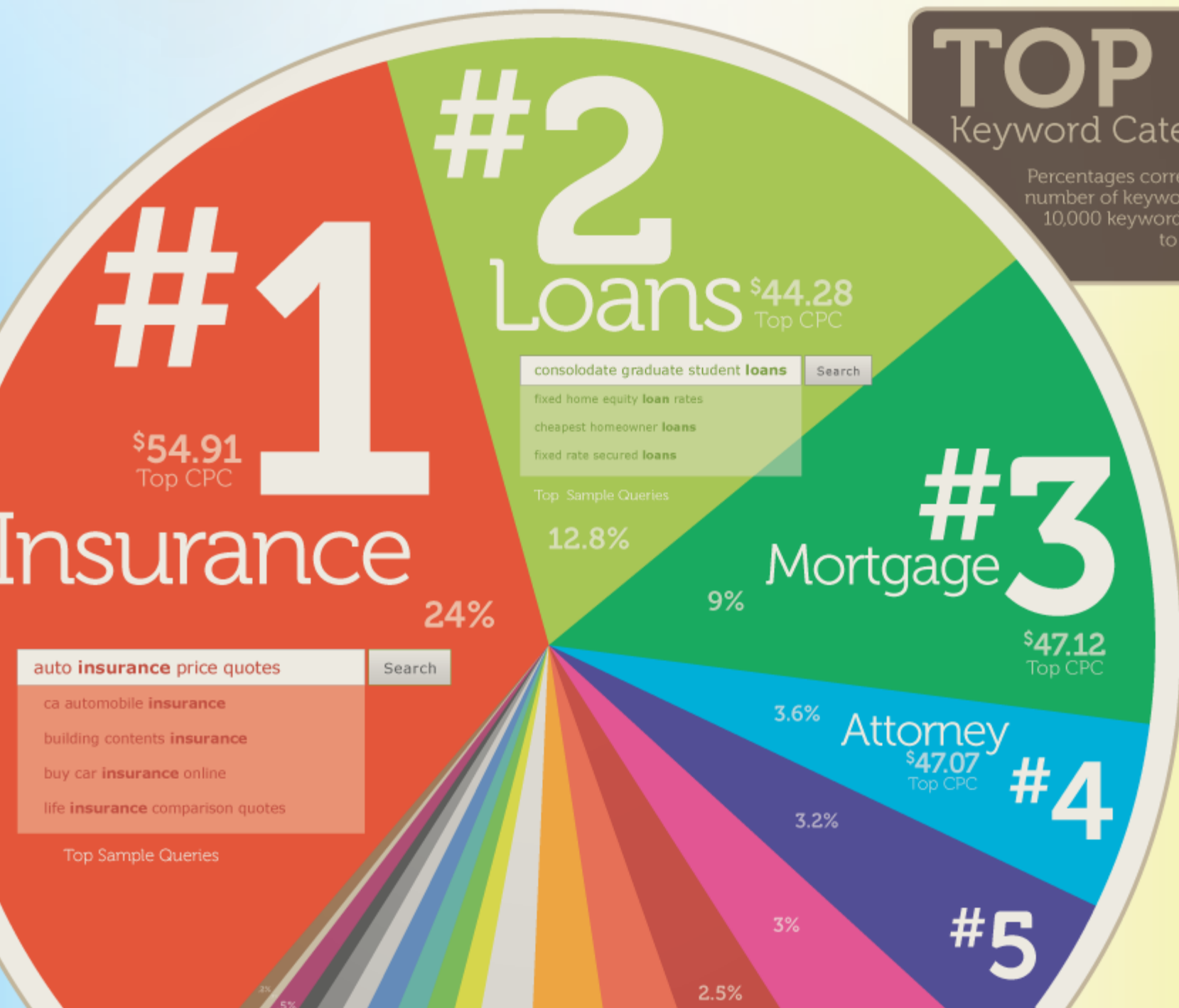


2002

TOP 20

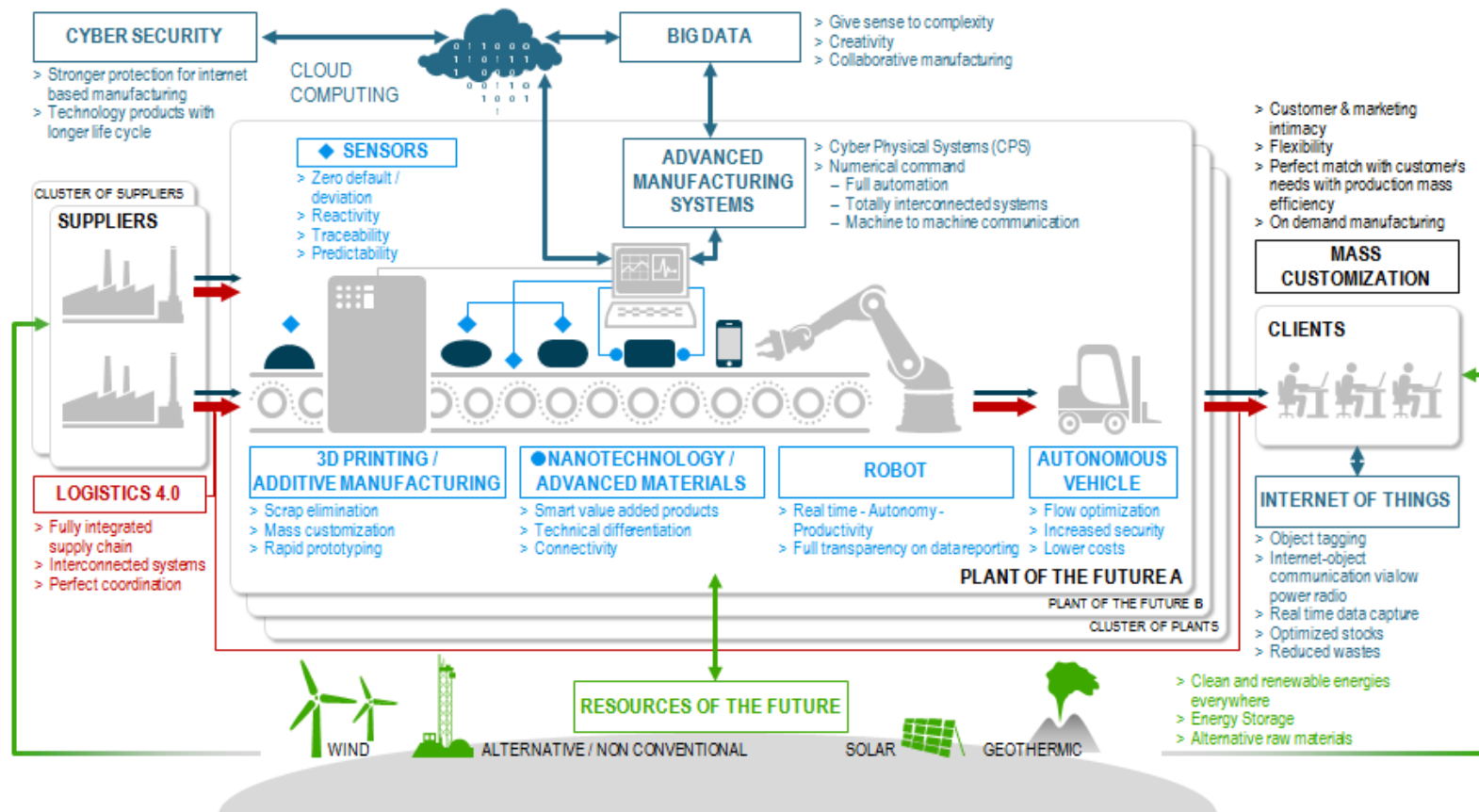
Keyword Categories

Percentages correspond to the number of keywords in the top 10,000 keywords that belong to that category.



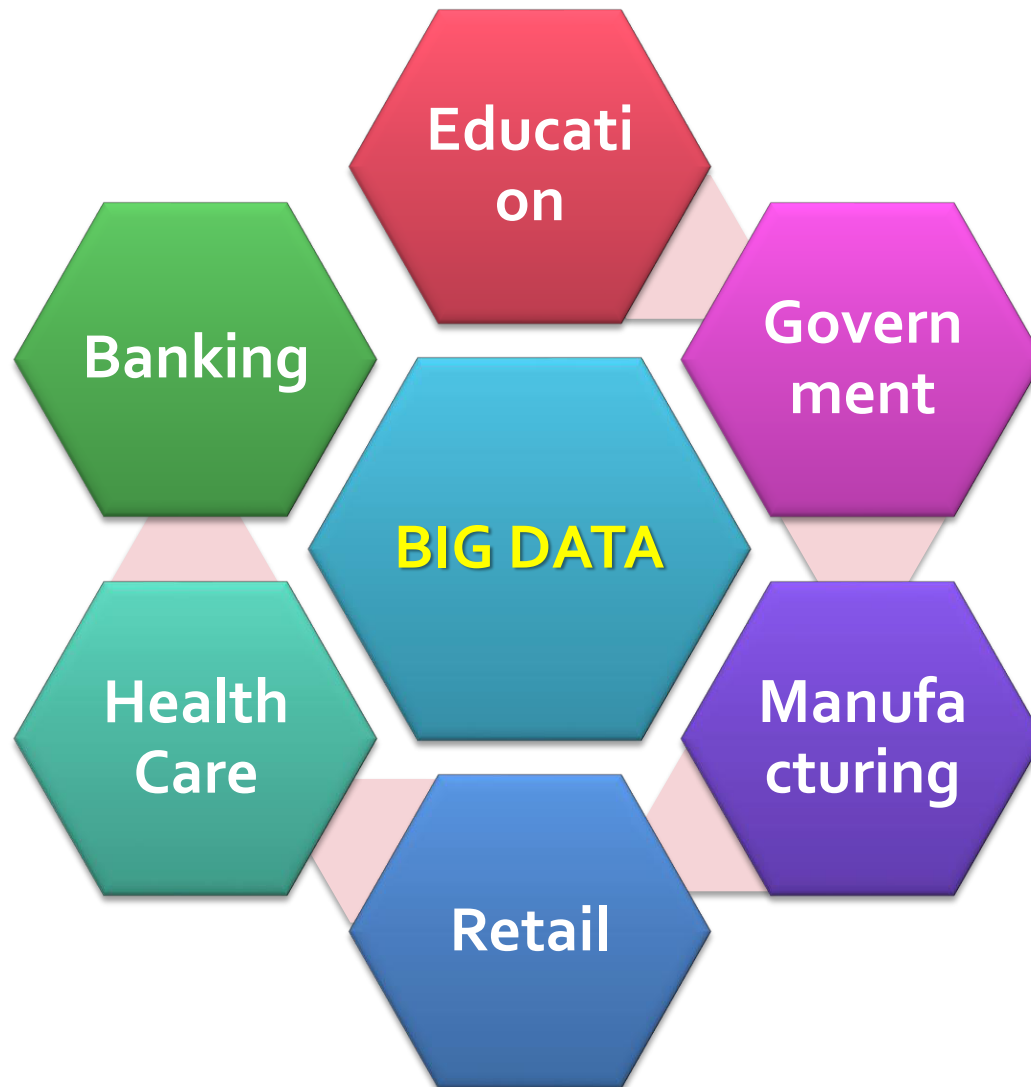
Big Data in Industry 4.0

The Industry 4.0 ecosystem

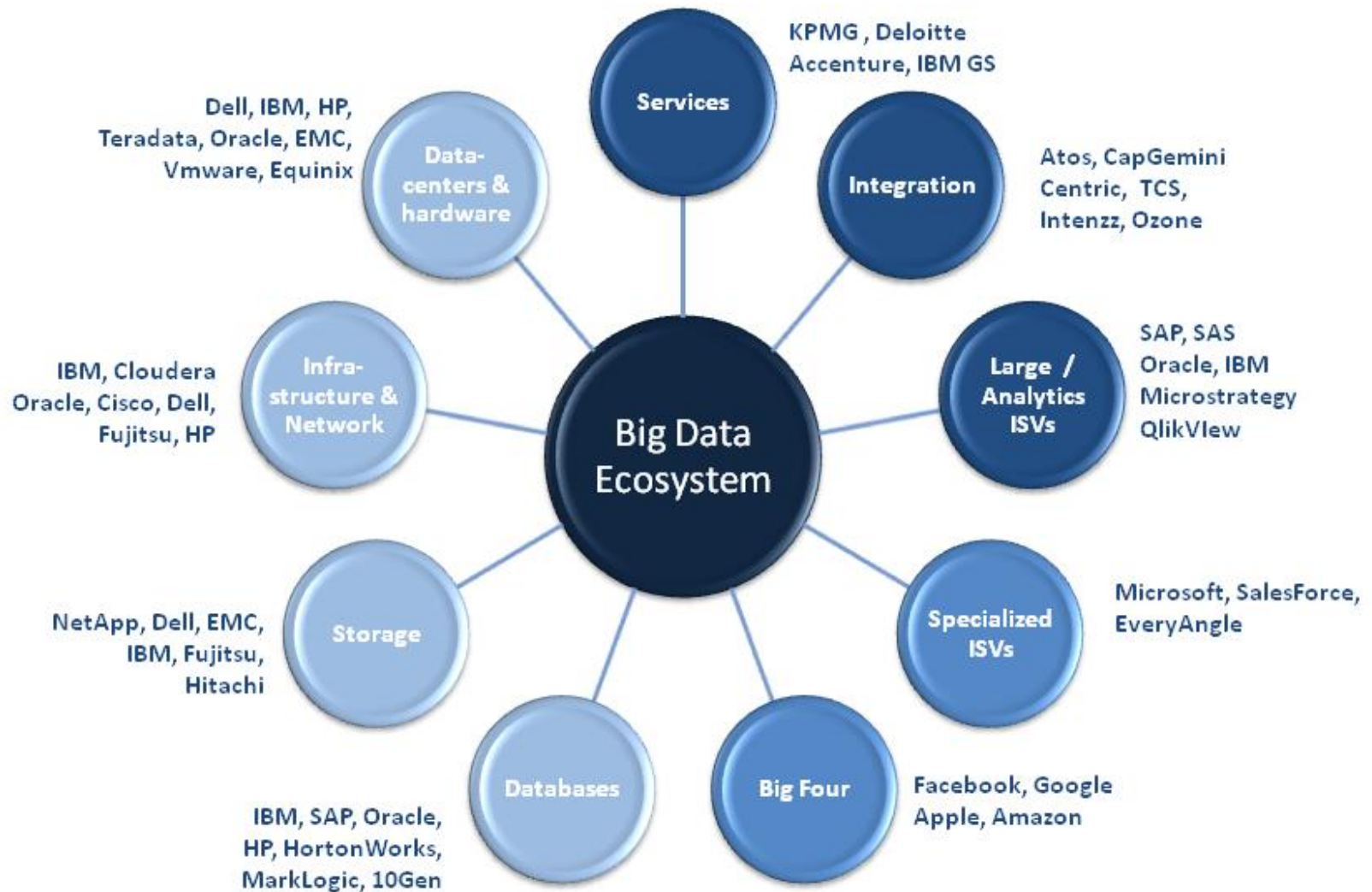


[source: Roland Berger]

BIG Data Users



BIG Data Ecosystem



Big Data Landscape

Vertical Apps



Log Data Apps



Ad/Media Apps



Business Intelligence



Analytics and Visualization



Data As A Service



Analytics Infrastructure



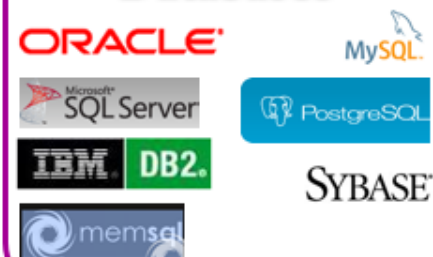
Operational Infrastructure



Infrastructure As A Service



Structured Databases



Technologies



DATA & AI LANDSCAPE 2020

INFRASTRUCTURE

STORAGE

HADOOP

DATA LAKES

DATA WAREHOUSES

STREAMING / IN-MEMORY

NoSQL DATABASES

NewSQL DATABASES

GRAPH DBs

MPP DBs

SERVER-LESS

CLUSTER SVCS

ETL / DATA TRANSFORMATION

DATA INTEGRATION

DATA GOVERNANCE

DATA QUALITY

MGMT / MONITORING

DATA GENERATION & LABELING

AI OPS

GPU DBs & CLOUD

AI HARDWARE

ANALYTICS & MACHINE INTELLIGENCE

BI PLATFORMS

VISUALIZATION

DATA ANALYST PLATFORMS

DATA SCIENCE NOTEBOOKS

DATA SCIENCE PLATFORMS

MACHINE LEARNING

COMPUTER VISION

HORIZONTAL AI

SPEECH & NLP

SEARCH

LOG ANALYTICS

SOCIAL ANALYTICS

WEB / MOBILE / COMMERCE ANALYTICS

APPLICATIONS - ENTERPRISE

SALES

MARKETING - B2B

MARKETING - B2C

CUSTOMER EXPERIENCE / SERVICE

HUMAN CAPITAL

LEGAL

REGTECH & COMPLIANCE

FINANCE

AUTOMATION & RPA

SECURITY

APPLICATIONS - INDUSTRY

ADVERTISING

EDUCATION

REAL ESTATE

GOVT & INTELLIGENCE

COMMERCE

FINANCE - LENDING

INSURANCE

HEALTHCARE

LIFE SCIENCES

TRANSPORTATION

AGRICULTURE

INDUSTRIAL

OTHER

OPEN SOURCE

FRAMEWORKS

QUERY / DATA FLOW

DATA ACCESS & DATABASES

ORCHESTRATION & PIPELINES

STREAMING & MESSAGING

STAT TOOLS & LANGUAGES

AI OPS & INFRA

AI / MACHINE LEARNING / DEEP LEARNING

SEARCH

LOGGING & MONITORING

VISUALIZATION

COLLABORATION

SECURITY

DATA SOURCES & APIs

DATA MARKETPLACES & DISCOVERY

FINANCIAL & ECONOMIC DATA

AIR / SPACE / SEA

PEOPLE / ENTITIES

LOCATION INTELLIGENCE

OTHER

DATA RESOURCES

DATA SERVICES

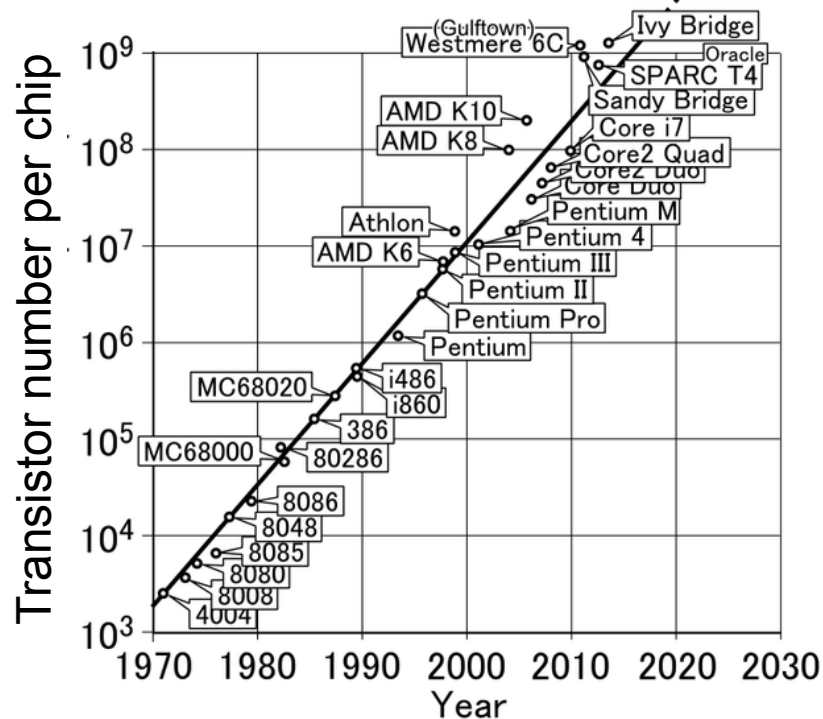
INCUBATORS & SCHOOLS

RESEARCH

Computing power

Scale-up

Moore's Law



(Graph from betanews)

Scale-out



Parallel and Distributed

Geo-distributed server farms



*Google server farms
2.5 million machines (est)*

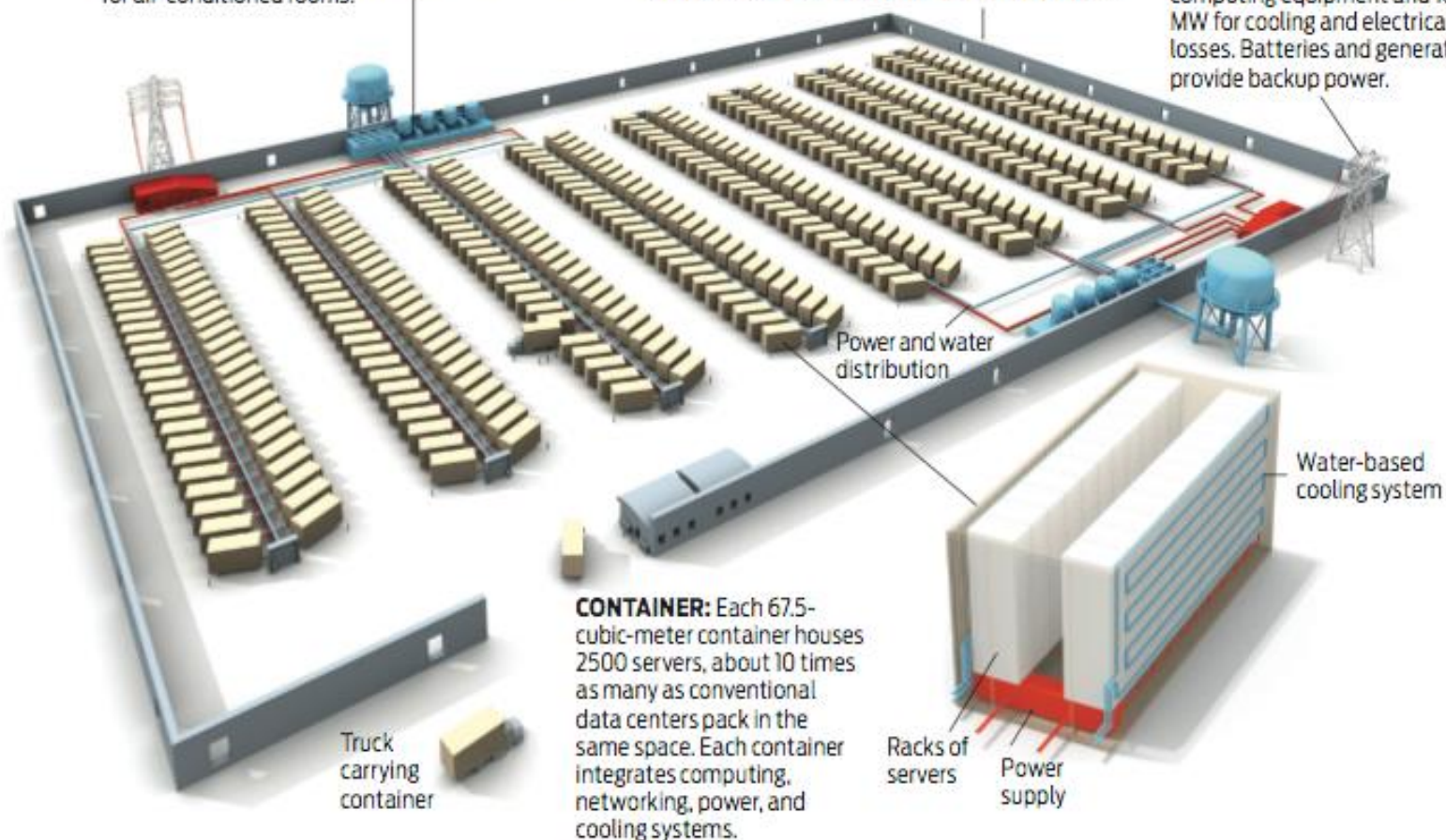


The Million-Server Data Center

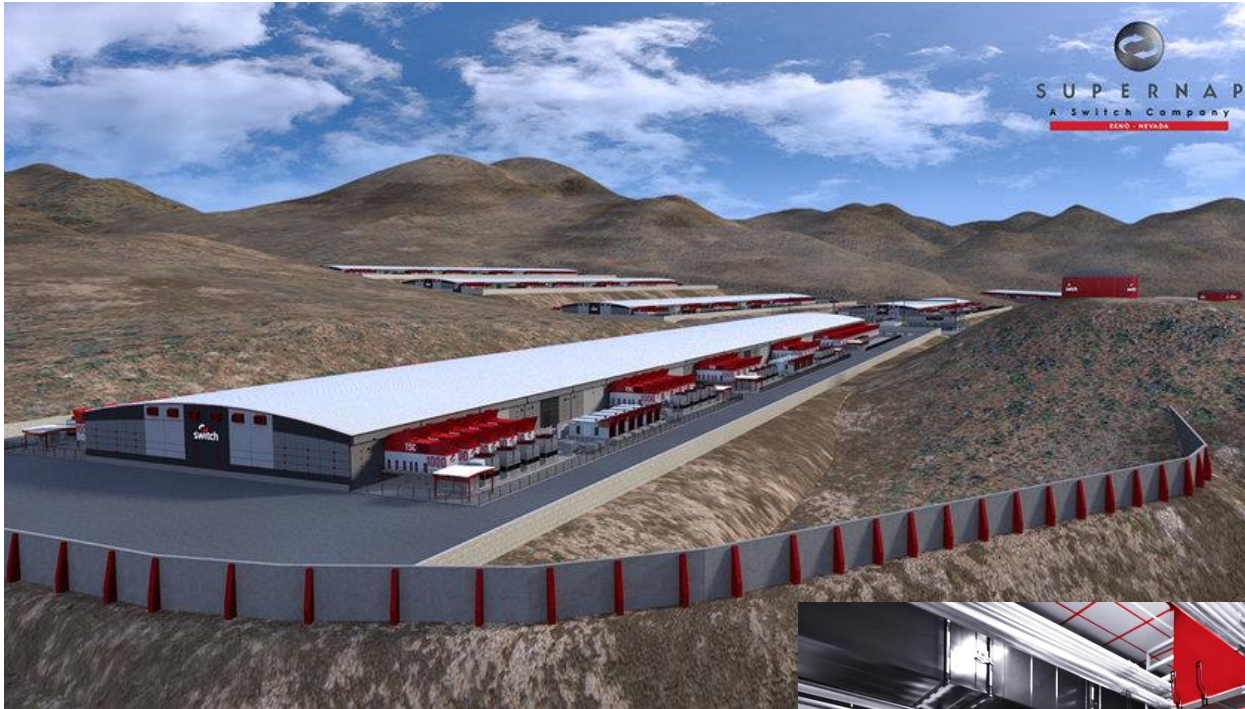
COOLING: High-efficiency water-based cooling systems—less energy-intensive than traditional chillers—circulate cold water through the containers to remove heat, eliminating the need for air-conditioned rooms.

STRUCTURE: A 24 000-square-meter facility houses 400 containers. Delivered by trucks, the containers attach to a spine infrastructure that feeds network connectivity, power, and water. The data center has no conventional raised floors.

POWER: Two power substations feed a total of 300 megawatts to the data center, with 200 MW used for computing equipment and 100 MW for cooling and electrical losses. Batteries and generators provide backup power.



Switch Data Center



Pronghorn of UNR is hosted in Switch Tahoe Reno Campus

<https://www.switch.com/locations/>



Cluster specs

CPU Node Count	108
GPU Node Count	11
Processor Count	3456
GPU Count	44
Memory	24.8 TB
Intranet	100Gb/s Intel Omni-Path
Internet	10Gb/s (upgrade to 100Gb/s)
Storage Capacity	1.5 PB
Theoretical Performance	310TFLOPS (30X The Grid)

CPU Node specs

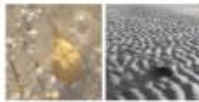
Processor Count	32
Processor Specs	Intel Xeon E5-2683 v4, 2.1GHz
Memory	256GB
Networking	100Gb/s Intel Omni-Path

GPU Node specs

Processor Count	32
Processor Specs	Intel Xeon E5-2683 v4, 2.1GHz
Memory	256GB
Networking	100Gb/s Intel Omni-Path
GPU Count	4
GPU Specs	NVIDIA Tesla P100, 16GB Memory, NVLink

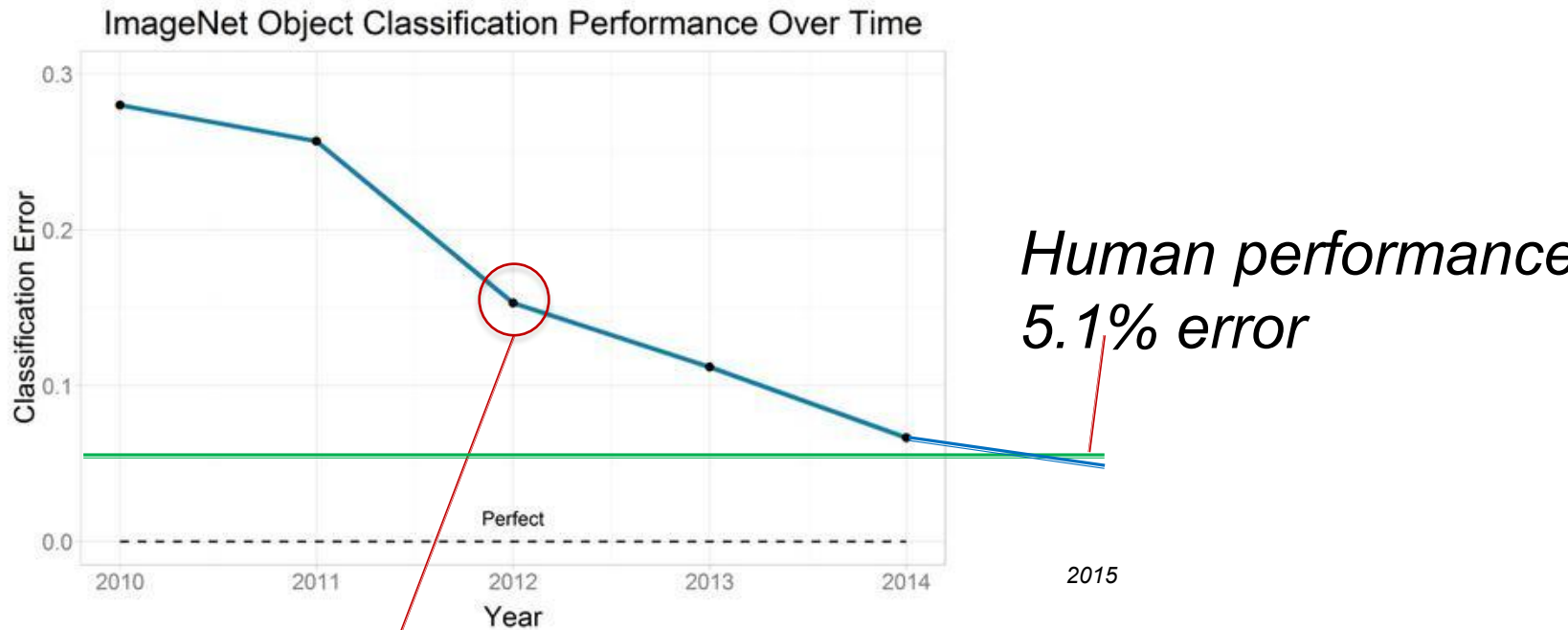
The unreasonable effectiveness of Deep Learning (CNNs)

2012 Imagenet challenge:
Classify 1 million images into 1000 classes.

 Cliff dwelling L2 11.0% - Mah. 99.9%	 horseshoe crab 0.99%	 African elephant 0.99%	 mongoose 0.94%	 Indian elephant 0.88%	 dingo 0.87%
	 cliff 0.07%	 dam 0.00%	 stone wall 0.00%	 brick 0.00%	 castle 0.00%
 Gondola L2 4.4% - Mah. 99.7%	 shopping cart 1.07%	 unicycle 0.84%	 covered wagon 0.83%	 garbage truck 0.79%	 forklift 0.78%
	 dock 0.11%	 canoe 0.03%	 fishing rod 0.01%	 bridge 0.01%	 boathouse 0.01%
 Palm L2 6.4% - Mah. 98.1%	 crane 0.87%	 stupa 0.83%	 roller coaster 0.79%	 bell core 0.78%	 flagpole 0.75%
	 cabbage tree 0.81%	 pine 0.30%	 pandanus 0.14%	 iron tree 0.07%	 logwood 0.06%

The unreasonable effectiveness of Deep Learning (CNNs)

Performance of deep learning systems over time:

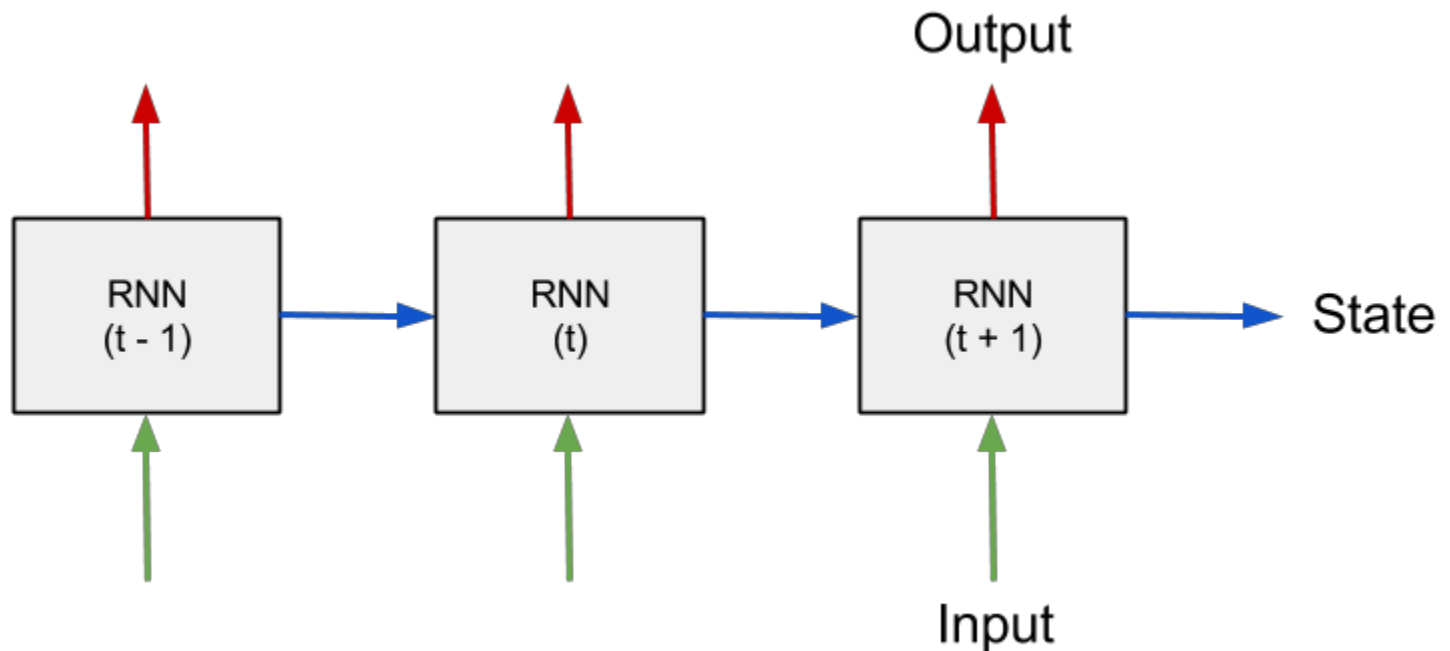


Krizhevsky, Sutskever, and Hinton, NIPS 2012

The unreasonable effectiveness of RNNs

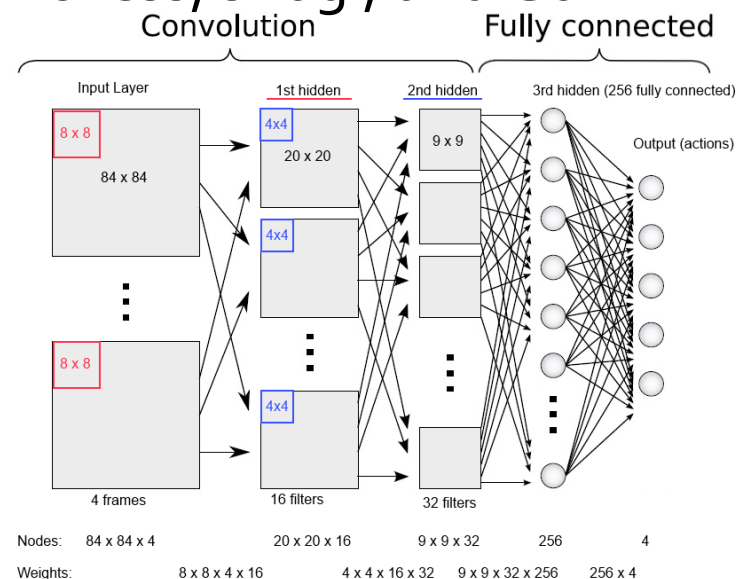
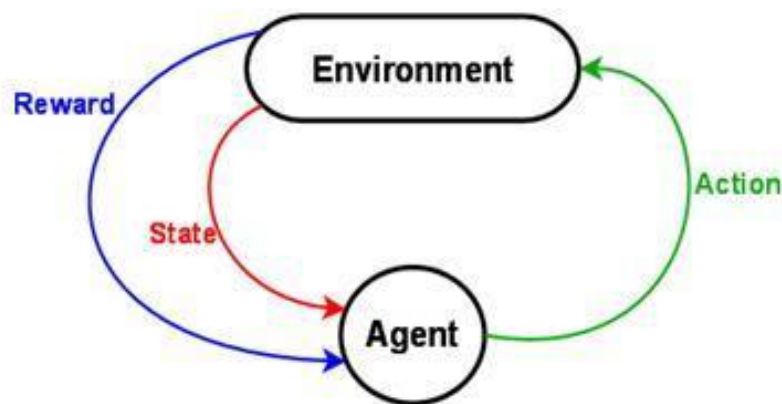
RNNs are Recurrent Neural Networks, and have shown dramatic improvements in text-related tasks:

- Image captioning
- Language translation
- Analogy and semantic queries



The unreasonable effectiveness of Reinforcement Learning + Deep Learning

- In 2013, Deep Mind demonstrated superior performance (better than a human expert) on six games on a virtual Atari 2600 game console (Pong, Breakout, Space Invaders...)
 - <https://www.youtube.com/watch?v=V1eYniJoRnk>
- Acquired by Google in 2014
- In 2017, Deep Mind demonstrated superior performance (better than a human expert) on chess, shogi, and Go.



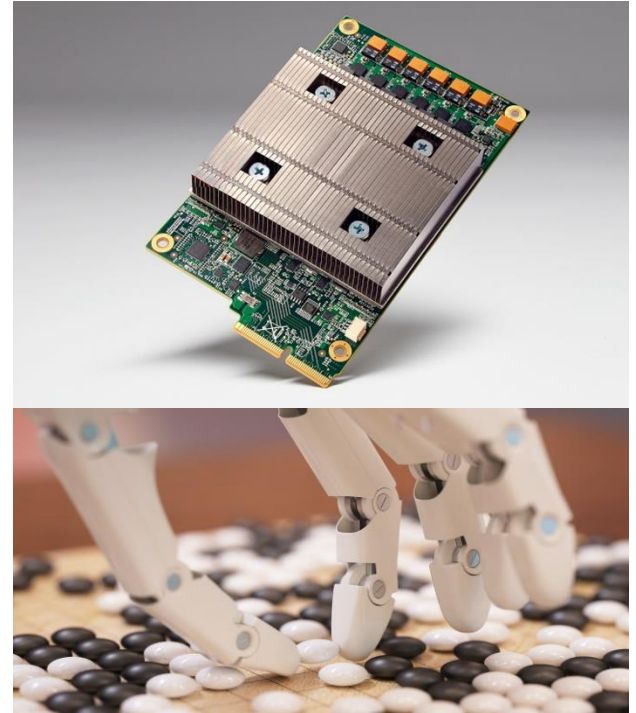
The unreasonable effectiveness of Reinforcement Learning + Deep Learning



Chess

IBM Deep Blue (1997)

VS



Go

Alpha Go (2017)

ChatGPT (Chat Generative Pre-trained Transformer)

- **ChatGPT** is a chatbot launched by OpenAI in November 2022. It is built on top of OpenAI's GPT-3 family of large language models, and is fine-tuned (an approach to transfer learning) with both **supervised** and **reinforcement learning** techniques.

Try it!

<https://openai.com/blog/chatgpt/>



To build a CNN model, you can use a deep learning framework such as TensorFlow or PyTorch. Here is an example of how to build a simple CNN model using TensorFlow:



1. Import the necessary libraries and modules:

Copy code

```
import tensorflow as tf
from tensorflow.keras import layers
```

2. Define the model architecture:

Copy code

```
model = tf.keras.Sequential()
model.add(layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(64, (3, 3), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(64, (3, 3), activation='relu'))
```

3. Add the fully connected layers:

Copy code

```
model.add(layers.Flatten())
model.add(layers.Dense(64, activation='relu'))
model.add(layers.Dense(10, activation='softmax'))
```

4. Compile the model:

Regenerate response

Copy code



Big Data Can Be Challenging



Google Flu Trends

- *Detecting outbreaks two weeks ahead of CDC data*
- *Poor prediction results*
- *Incorporating CDC data*
- *Flue Trends (healthmap.org)*



Risks of Big Data

- Short of well-trained experts
 - Need the right people to solve the right problems
- Bad data, analytics
 - Danger of black-box approaches
- Costs escalate too fast
 - Expensive to store and analyze
 - Isn't necessary to capture 100%
- Security & privacy
 - Data theft
 - Lots of data is sensitive in nature
 - Self-regulation & legal regulation



Consumption	CO ₂ e (lbs)
Air travel, 1 person, NY↔SF	1984
Human life, avg, 1 year	11,023
American life, avg, 1 year	36,156
Car, avg incl. fuel, 1 lifetime	126,000

Training one model (GPU)	
NLP pipeline (parsing, SRL)	39
w/ tuning & experiments	78,468
Transformer (big)	192
w/ neural arch. search	626,155



Emma et al "Energy and Policy Considerations for Deep Learning in NLP".



Why You Need to Tame Big Data

- Analyzing big data is already a norm (e.g., business, health, science)
- Be left behind in a few years
 - So far, only missed the chance on the bleeding edge
- Capturing data, using analysis to make decisions
 - Just an extension of what you are already doing today

The Model Has Changed...

- The Model of Generating/Consuming Data has Changed

Old Model: Few companies are generating data, all others are consuming data



New Model: all of us are generating data, and all of us are consuming data



The Model Has Changed...



< The past >



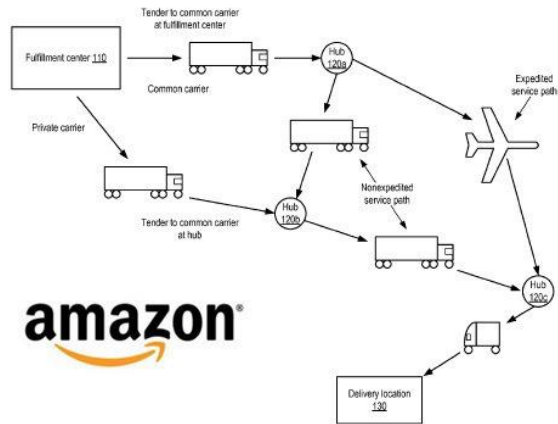
< The present >



< The future >

The Model Has Changed...

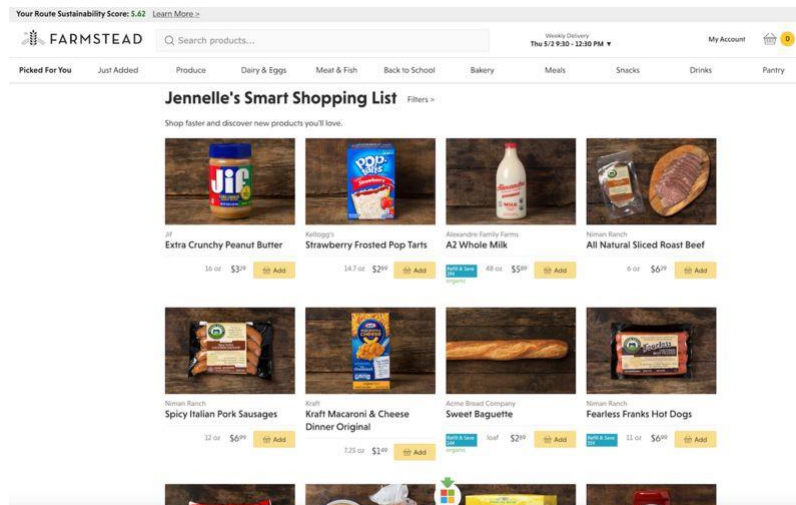
Amazon Predictive Shopping/Anticipatory Shipping



Samsung Smart Refrigerator



Predictive Shopping List



Future of Big Data



- Intelligence – understand & interact
 - Machine learning, deep learning
 - Devices better understand & interact with physical world
- Cloud, serverless computing
 - On-premises, fragmental -> remote, consolidated
 - Efficient, robust, scalable, convenient
- Big data at personal scale
 - Free from non-creative thinking tasks
 - Health monitoring
 - Data can represent “you”



THANK YOU!

Questions?